BFW STEAM HEATER VER 1

Operator Training Simulator · E-101

Chulalongkorn University · Department of Chemical Engineering

PROCUREMENT, COMPLETION AND COMMISSIONING PLAN

DOCUMENT NO. : BFWH-V1-CM-PLN-001

DISCIPLINE : Commissioning

PROJECT : BFW Steam Heater Ver 1 · E-101 Operator Training Simulator

STATUS : Issued for teaching (IFT)

Plan from purchase to handover of the BFW steam heater unit E-101: 12 procurement packages (4 long-lead), the inspection and test plan with E-101 hydrotests (shell 455, tube 1,430 kPa g), SIS proof tests before start-up, pre-commissioning checklists per system and the first start-up mapped to the cold start-up procedure.

Rev	Date	Description	Originator	Checker	Approver
0	25-Oct-26	Issued for teaching	A. Soottitantawat	—	—

Numbers are taken from the BFW Steam Heater Ver 1 simulator model; values marked assumed / TBC / HOLD are not design data.

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1 Phases after design

Each phase ends with a certificate the next relies on, and the work is organised by system, the unit that can be tested and handed over on its own.

Phase	What happens	Who leads	Ends with
Procurement	Buy the equipment and bulk to the data sheets; expedite long-lead items; inspect at the vendor (FAT).	Procurement, with the discipline engineers	Purchase orders, vendor data, FAT reports
Construction	Install, erect and connect: foundations, equipment setting, piping, cabling, insulation.	Construction contractor	Installed plant, test packs
Mechanical completion (MC)	Everything installed as per the drawings and tested static: hydrotests, continuity, punch list A cleared.	Construction, accepted by Commissioning	MC certificate per system
Pre-commissioning	Systems made ready without process fluid at operating conditions: flushing, steam blowing, loop checks, stroke tests, motor runs, SIS proof tests.	Commissioning team	Checklists signed per system; ready-for-commissioning (RFC)
Commissioning	Systems brought into service with the real fluids at low duty: water circulation, first steam, controller tuning.	Commissioning team with Operations	Systems in service
Start-up	Feed-in and line-out at design: the cold start-up procedure, performance test.	Operations, supported by Commissioning	Performance test record
Handover	Unit, documents and spares handed to the owner; punch list B agreed; as-built documents.	Project to Operations / Maintenance	Handover certificate, as-built dossier

2 Procurement packages

Lead times and inspection levels are assumed, typical values. Inspection levels: Level 1 — full: inspector at key hold points and the FAT; Level 2 — witness the FAT and final inspection; Level 3 — review certificates, final inspection; Level 4 — certificates only.

Pkg	Title	Items	Specified by	Long lead	Weeks (assumed)	Insp. level	Key inspections and tests
PK-01	BFW heater E-101	Shell-and-tube heater, 100 tubes 15.875 × 1.651 mm, 1 pass; shell and tube nozzles, VB-101 and vent / drain connections	Mechanical data sheet (Mechanical tab), Design tab sizing	Yes	20–26	1	Material certificates, weld procedure and NDT review, hydrotest witness (shell and tube), dimensional check, nameplate
PK-02	BFW pumps P-101A/B with motors	Two centrifugal pumps, motors 37 kW (Electrical load list), base plates, couplings, seals	Pump data sheet (Mechanical), BFWH-V1-EE-LL-001	Yes	16–20	2	Performance test (Q–H, power, NPSHr) witnessed, mechanical run test, hydrotest of casing
PK-03	Control valves	FV-101A (fail open), FV-101B (minimum-flow recycle, fail position TBC), FV-104 (fail closed), positioners	Instrument data sheets (Instrumentation tab)	—	12–16	2	Stroke and fail-action test, seat-leakage test, body hydrotest, positioner calibration
PK-04	SIS shut-off valve XV-100	XV-100 with spring-return actuator, solenoid (SOV), limit switches; fail closed	SRS BFWH-V1-IN-SRS-001, C&E BFWH-V1-IN-CE-001	Yes	16–20	1	Stroke time (≤ 2 s assumed in the SRS), close on de-energise and on loss of air, seat-leakage class, SIL certificate of the assembly
PK-05	Relief and vacuum devices	PSV-100 (set TBC), RD-100 (HOLD, tube-rupture case), VB-101 vacuum breaker	Relief & vent tab	—	10–14	2	PSV bench set test and seat tightness; certified capacity; RD burst certificate (when released)
PK-06	Steam trap set and steam specialities	Trap ST-101, strainer Y-101, rev B RO-101, rev C AAV-101 with TI-102, drip-leg trap upstream of XV-100	Process / Mechanical (trap capacity from the Design tab)	—	6–10	3	Trap capacity certificate at low differential pressure, model and orientation
PK-07	Field instruments	FE/FT-101, FE/FT-104, TT-101, TT-108, PT-100, PDT-109, TI-100, TI-102, TI-103, PI-105, PI-106, gauges	Instrument data sheets and index (Instrumentation tab)	—	8–12	3	Calibration certificates, orifice plate dimensional check, SIL capability of FT-101 / TT-108 / PT-100
PK-08	DCS and SIS I-100	Controllers, I/O, operator station, SIS logic solver with C&E logic	C&E BFWH-V1-IN-CE-001, SRS BFWH-V1-IN-SRS-001	Yes	20–24	1	Integrated FAT: every C&E cause and effect, alarms, start permissives, first-out, bypass logging
PK-09	BFW tank TK-101	Atmospheric BFW storage tank with nozzles (scope assumed in this project)	Mechanical data sheet	—	12–16	2	Material certificates, weld NDT, water-fill test, nozzle orientation
PK-10	Piping and bulk	Pipe, fittings, flanges, gaskets, bolting, supports; manual valves V-101…V-115, NRV-101/102 (lines 8"-BW-100 … 4"-SC-102)	P&ID BFWH-V1-PID-001 rev D, line list (Process)	—	6–10	3	Material certificates, valve pressure tests, check-valve direction (HAZOP R04)
PK-11	Insulation and personnel protection	Hot insulation of shell, steam and condensate lines; guards on hot surfaces; vent silencer	Insulation specification (Mechanical)	—	4–8	4	Material certificates, thickness check on site
PK-12	Electrical bulk	MCC starters, power and control cables, lighting, UPS feeders	BFWH-V1-EE-LL-001	—	10–14	3	Routine tests of MCC at the works; cable drum certificates

3 Inspection and testing

Hydrotest of E-101 at 1.3 × the design pressure (assumed code factor, no temperature correction), using the Design-tab design pressures: shell 1.3 × 350 = **455 kPa g**, tube side 1.3 × 1,100 = **1,430 kPa g**. The shell design pressure is not yet agreed between documents (LOPA / Safety layers basis 300 kPa g → 390 kPa g): the shell test pressure is HOLD (EQ-60, EQ-01).

Item	Test	Acceptance	Where	Witness
E-101 shell side	Hydrotest at 1.3 × design = 1.3 × 350 = 455 kPa g	No leak, no permanent deformation; hold time per code (assumed 30 min)	Vendor works (FAT)	Inspector witness
E-101 tube side	Hydrotest at 1.3 × design = 1.3 × 1100 = 1430 kPa g	No leak at tube-to-tubesheet joints (shell open / drained to see)	Vendor works (FAT)	Inspector witness
Piping test packs	Hydrotest of each test pack at 1.5 × line design pressure (assumed, piping code) after welding and NDT	No leak; test records	Site	Construction QC, Commissioning review
P-101A/B	Performance test on the vendor test bed; run test on site uncoupled then coupled	Head and power within vendor tolerance; vibration and bearing temperature within limits	FAT + site	Witness at FAT
FV-101A, FV-101B, FV-104	Full stroke and fail-action test; seat leakage	Smooth stroke; fail positions as the C&E (FV-101B TBC)	FAT + site	Witness
XV-100 + SOV	Stroke time and close on de-energise / loss of air; seat leakage	Closes in ≤ 2 s (assumed in the SRS)	FAT + site (SAT)	Witness
PSV-100	Bench test at the set pressure (set TBC) and seat tightness; seal	Opens within ±3 % of set (assumed), tight at 90 % of set	Site workshop before installation	Inspector witness
VB-101	Function test: opens at a small vacuum, closes at positive pressure	Opens at the specified vacuum (data sheet)	Site	Commissioning
Instrument loops	Loop check from field device to DCS display and alarm (inject 0/50/100 %)	Reading, range and alarm set points as the instrument index and ALM list	Site	Commissioning + Operations
DCS and SIS I-100	Integrated FAT; SAT after installation	Every C&E cause gives every effect; start permissives	FAT + SAT	Instrument + Process safety
SIS proof test before start-up	Each SIF end to end: sensor → logic → XV-100 closed, FIC-104 MAN 0 %, TIC-101 MAN	Trip at the set point after the delay; first-out correct; reset only when the cause is gone	Site, before the first steam	Process safety witness

SIS proof tests before start-up (each SIF end to end; annual thereafter, SRS)

SIF	Cause	Set point	Delay s	Proof test
FSLL-101	Low-low BFW flow (armed in RUN)	< 30 m³/h	5	Simulate FT-101 below the set point with the SIS in RUN (XV-100 open, no steam): XV-100 must close after the delay; check it does not act in IDLE.
TSHH-108	High BFW outlet temperature	> 95 °C	5	Heat TE-108 in a calibration bath (or inject at the transmitter) above the set point: XV-100 closes after the delay.
PSHH-100	High shell pressure	> 250 kPa g	3	Pressurise PT-100 with a hand pump above the set point: XV-100 closes after the delay.
HS-ESD	Manual emergency shutdown	push-button	0	Press the ESD button: XV-100 closes immediately; same effects as any trip.

Expected effect of every SIF: XV-100 closed, FIC-104 in MAN at 0 %, TIC-101 in MAN, first-out shown; BFW pumps keep running.

4 Pre-commissioning checklists

Signed per system; the system is ready for commissioning (RFC) when every item is signed.

System	Item	Done
BFW system (8"-BW-100, 6"-BW-101, 3"-BW-102, 6"-BW-103)		
	Test packs hydrotested and signed; temporary strainers fitted at the P-101A/B suctions	☐
	BFW lines flushed with E-101 bypassed by a temporary spool, at not less than the design velocity, until the temporary strainers stay clean on two checks (assumed)	☐
	E-101 tube side flushed last; spool removed, permanent spools reinstated with new gaskets	☐
	Check valves NRV-101/102 and the BFW non-return to the boiler house fitted in the flow direction (HAZOP R04)	☐
	P-101A/B: alignment, lubrication, seal flush, rotation checked uncoupled; first run on recycle	☐
Steam supply (12"-LS-101, 14"-LS-102)		
	Drip leg and trap upstream of XV-100 installed (HAZOP R26)	☐
	FE-104, FV-104 and XV-100 removed or spooled out; temporary exhaust to a safe place with a silencer	☐
	Steam blow of LS-101 / LS-102 with heat-up and cool-down cycles	☐
	Target plates: two consecutive blows with no raised impact larger than 0.5 mm and no more than a few small pits (acceptance assumed; the project steam-blow procedure governs, EQ-63)	☐
	FE-104, FV-104, XV-100 reinstated with new gaskets; the steam branch kept isolated	☐
Condensate and trap (3"-SC-101, 4"-SC-102)		
	Condensate lines flushed to the return header	☐
	Trap ST-101 model and orientation as the data sheet; capacity ≥ 24,838 kg/h at start-up (Design tab, safety factor 2.5)	☐
	Strainer Y-101 screen fitted and clean; to be cleaned again after the first week (HAZOP R15)	☐
	Trap bypass V-103 closed; V-101 and V-102 open (start permissive)	☐
Shell vents and relief		
	Air vent V-104 line routed to a safe place with silencer; signs and hearing-protection zone (HAZOP R17)	☐
	VB-101 vacuum breaker function tested (HAZOP R14)	☐
	PSV-100 bench tested at its set point and sealed — set point TBC; RD-100 HOLD (EQ-61)	☐
	Rev B RO-101 or rev C AAV-101 with TI-102 checked if fitted	☐
Instruments, DCS and SIS		
	Loop check of FT-101, FT-104, TT-101, TT-108, PT-100, PDT-109, TI-100, TI-103, PI-105, PI-106	☐
	Alarm set points loaded as the alarm list (FIC-101 80/120 m³/h, TIC-101 67/74 °C, TIAH-108 85 °C, PI-100 H 200 kPa g …)	☐
	SIS trip set points as the C&E: FSLL-101 < 30 m³/h 5 s, TSHH-108 > 95 °C 5 s, PSHH-100 > 250 kPa g 3 s	☐
	SIS proof test of each SIF before the first steam (section 3) and ESD push-button test	☐
	Start permissives tested: SIS IDLE, FIC-101 ≥ 50 m³/h, FIC-104 in MAN ≤ 10 %, V-101 and V-102 open	☐
Valves: stroke tests		
	FV-101A full stroke; fails open on loss of air	☐
	FV-101B full stroke; follows 100 − FIC-101 OUT; record its fail position (TBC, EQ-03)	☐
	FV-104 full stroke; fails closed on loss of air	☐
	XV-100 stroke time ≤ 2 s (assumed in the SRS, EQ-62); closes on de-energise and on loss of air (HAZOP R22)	☐
Electrical		
	Motor insulation resistance and rotation of P-101A/B; MCC starter and local stop tested	☐
	UPS supply to the DCS, SIS and XV-100 solenoid tested on mains failure	☐
	Area lighting and socket outlets energised	☐

5 Commissioning and first start-up

The first start-up follows the operating procedure (SOP-E101-01 pre-start checks, SOP-E101-02 cold start-up, SOP tab of the simulator); the commissioning team adds what must be verified the first time.

Step	Cold start-up step	Commissioning: verify the first time	HAZOP
1	FIC-101 in MAN, open FV-101A a little and start P-101A	First run of P-101A on the process: discharge pressure PI-106 rises toward shut-off and falls as FV-101A opens; vibration and bearing temperature logged over the first hours. Test the start of P-101B as well.	R01, R03, R04
2	Bring BFW to 100 m³/h and put FIC-101 in AUTO at 100	Tune FIC-101 from the model starting values (Kc 0.4, Ti 5 s); confirm FT-101 against the pump curve (PI-106 ≈ 662 kPa g at 110.7 m³/h total flow).	R18
3	Line up the trap (V-101, V-102 open) and open air vent V-104	First line-up of the trap set; Y-101 clean; vent discharge routed safely; operators shown the vent and trap set.	R15, R17
4	FIC-104 in MAN at 5–10 % and open XV-100 from the SIS screen	First opening of XV-100 after the proof test; start permissives seen to work live; steam branch warmed with its drains open first.	R18, R26
5	Warm the shell for 2–3 minutes at low steam (no water hammer)	Hold point: listen for water hammer; watch shell and nozzle expansion, sliding saddle free; hot bolting of flanges at first heat-up (assumed practice).	R18
6	Raise the steam by hand with V-104 open until dry steam blows (air purged)	First air purge of a new shell takes longer (shell full of air); check Y-101 afterwards for construction debris.	R13, R18
7	Close V-104 (a crack of 10 % or less is allowed)	Set the continuous vent crack, or commission RO-101 / AAV-101 if fitted.	R13
8	FIC-104 to AUTO, then CAS; TIC-101 to AUTO at 70 °C	Tune FIC-104 (Kc 0.25, Ti 4 s) and TIC-101 (Kc 1.2, Ti 45 s) from the model starting values with small step tests; check the cascade.	R21
9	Line out: TT-101 70 ±1 °C in cascade with FV-104 below 90 %	Performance test at line-out: duty, steam flow, shell pressure, tube ΔP against the design (Q 6,040 kW, steam 9,935 kg/h, shell 105 kPa g, PDI-109 60 kPa): the clean baseline for fouling monitoring; then handover.	R07

HAZOP recommendations closed during commissioning

Rec.	Recommendation (HAZOP report BFWH-V1-SA-HZP-001)	Commissioning action	Where
R04	Confirm a check valve on each pump discharge and on the BFW line to the boiler house; add them to the inspection plan.	Check valves fitted in the right direction; added to the inspection plan	BFW system checklist; PK-10 inspection
R14	Confirm the shell is rated for full vacuum or that VB-101 is sized for the fastest condensation after a trip; add VB-101 to the preventive-maintenance and test plan.	VB-101 function test before start-up; in the preventive-maintenance plan	Shell vents checklist
R15	Trap management: regular trap survey (ultrasonic or temperature), Y-101 cleaning after any maintenance, and confirm ST-101 capacity at low shell differential pressure (start-up and near-vacuum operation).	Trap capacity confirmed; Y-101 cleaned after the steam blow and the first week	Condensate checklist
R18	Start-up procedure hold points: BFW in AUTO first, vent V-104 open, warm-up with FIC-104 below 10 % (the model gives water hammer at 10 % in a cold shell, none at 5 %) until the shell is warm, vent until dry steam blows, then cascade.	Start-up hold points written into the commissioning procedure	Start-up steps 2, 4, 5, 6
R19	Shutdown procedure: steam off and shell depressured before the BFW is stopped or blocked in; open V-105 only after the shell pressure has fallen; keep V-104 open during lay-up.	First normal shutdown done by procedure: steam off before BFW stopped	Commissioning: first shutdown (SOP-E101-03)
R20	Trip-bypass management for FSLL-101, TSHH-108 and PSHH-100: permit, time limit, MOC, bypass status shown on the DCS; proof-test the XV-100 trips at the interval set by LOPA.	SIS proof test before start-up; bypass register opened at handover	Section 3, SIS proof tests
R21	Restart after any trip: re-vent the shell (air drawn in through VB-101) before the cascade is put back; consider "V-104 open" as a start permissive.	Restart after the first trip re-vents the shell before the cascade	Start-up step 8; SOP-E101-05

Rec.	Recommendation (HAZOP report BFWH-V1-SA-HZP-001)	Commissioning action	Where
R22	Confirm the fail position of FV-101B (not shown on the P&ID) and that XV-100 closes on loss of air or power; document the fail positions in the cause-and-effect chart.	Fail positions of FV-101B and XV-100 confirmed by test and written in the C&E	Valve stroke-test checklist
R26	Confirm a drip leg and trap on the steam branch upstream of XV-100, and warm the line with its drains open before admitting steam.	Drip leg and trap upstream of XV-100 installed; line warmed with drains open	Steam supply checklist; start-up step 4

6 Open engineering queries

No.	Subject	Query	Ref.	Status
EQ-60	E-101 shell hydrotest pressure	The shell hydrotest is 1.3 × design: 455 kPa g with the Design-tab shell design pressure (350 kPa g) but 390 kPa g with the 300 kPa g basis of the LOPA / Safety layers tab. The test pressure cannot be released to the vendor until the shell design pressure is agreed (see EQ-01).	BFWH-V1-CM-PLN-001	Open
EQ-61	PSV-100 bench test and RD-100	PSV-100 cannot be bench-tested and sealed while its set point is TBC, and the tube-rupture relief device RD-100 is HOLD. Mechanical completion of the shell-relief system needs both; the relief package PK-05 is on the critical path.	BFWH-V1-PID-001	Open
EQ-62	XV-100 stroke time	The SRS response-time budget uses an XV-100 stroke of 2 s (assumed, TBC by vendor). Make the 2 s closing time a FAT acceptance criterion of PK-04 and repeat it at the SAT and at each proof test.	BFWH-V1-IN-SRS-001	Open
EQ-63	Steam-blow acceptance and temporary exhaust	The steam-blow target-plate acceptance and the temporary exhaust route for LS-101 / LS-102 are assumed in this plan. Agree the criteria with the owner and the noise / exclusion zone for the blow before construction ends.	BFWH-V1-CM-PLN-001	Open

Notes

- Model numbers (trip set points, start permissives, design point, trap requirement, design pressures) come from the simulator through epc/tools/ecomm_data.py; the Procurement to commissioning tab checks them in the page.
- Lead times, inspection levels, hold times, steam-blow acceptance and the hydrotest factor are teaching assumptions.